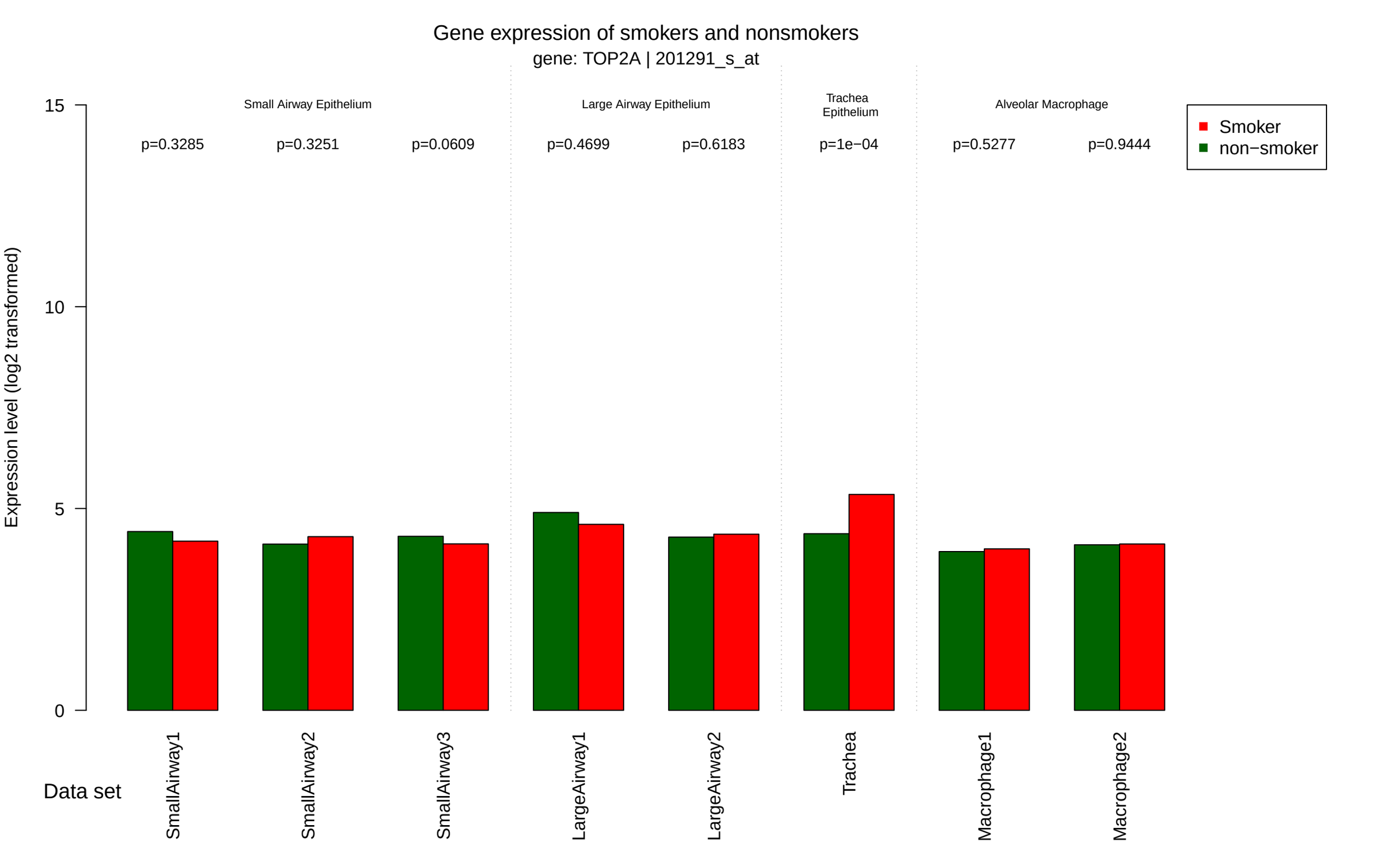
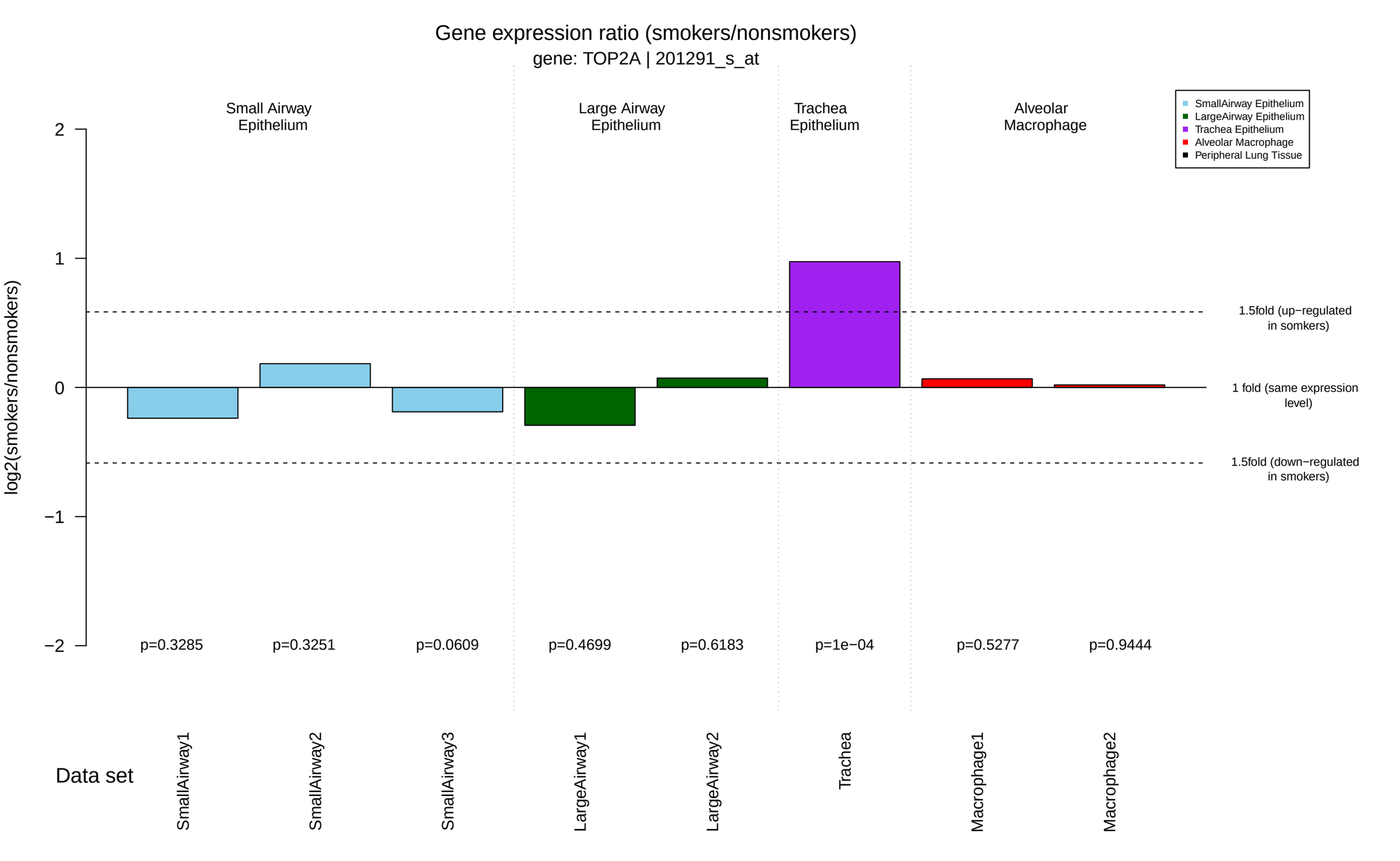
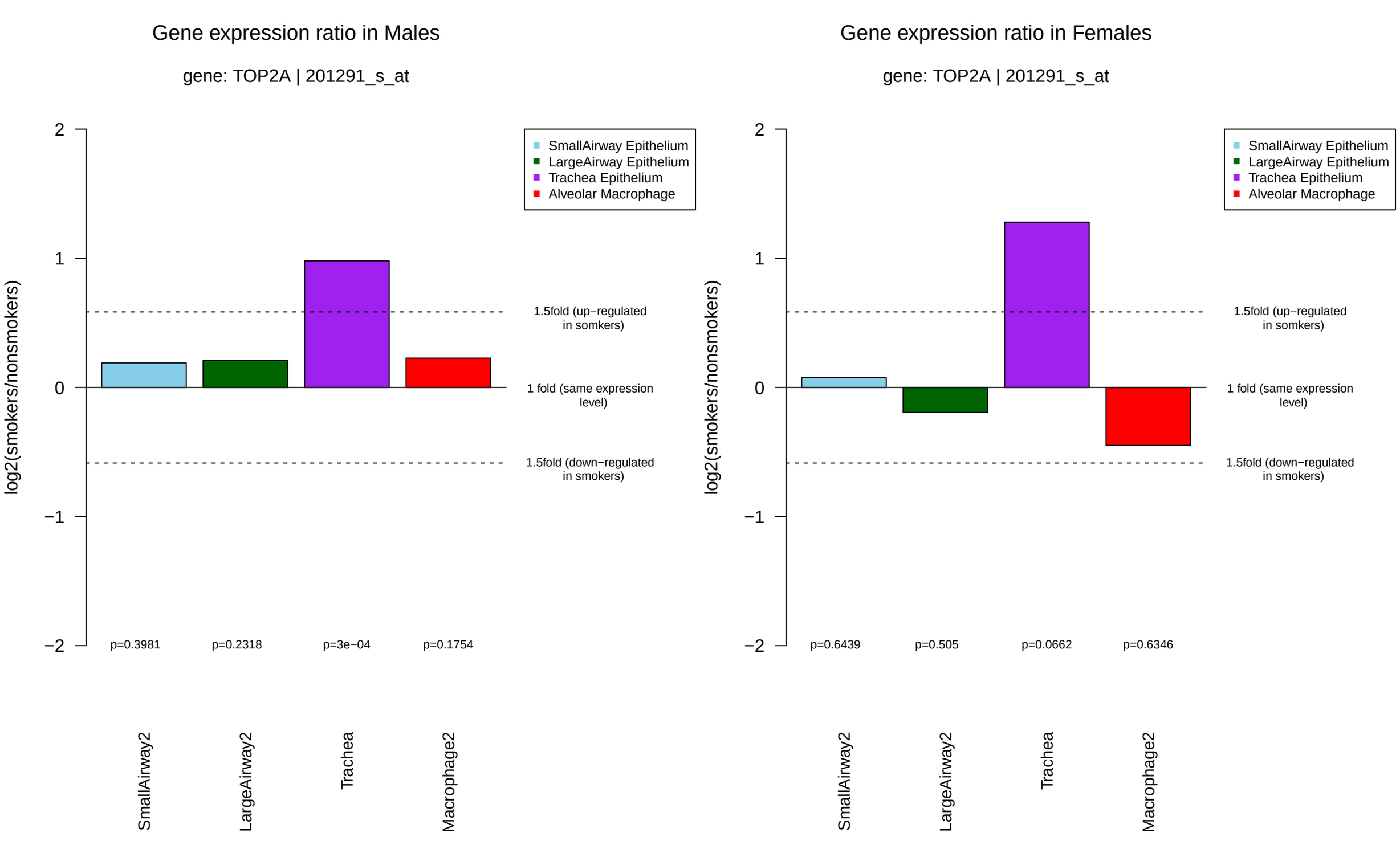
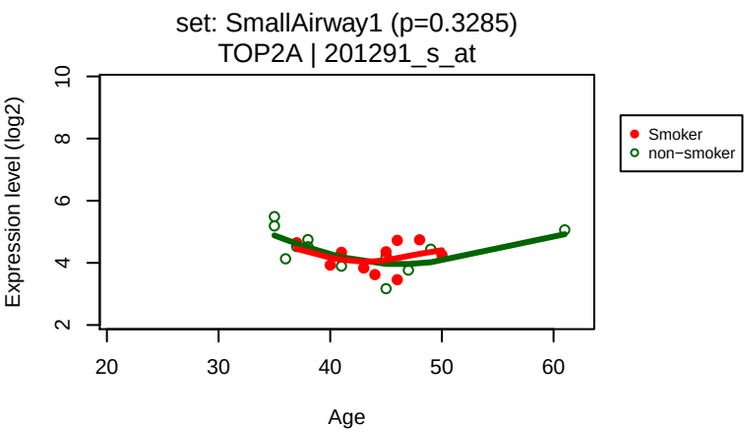
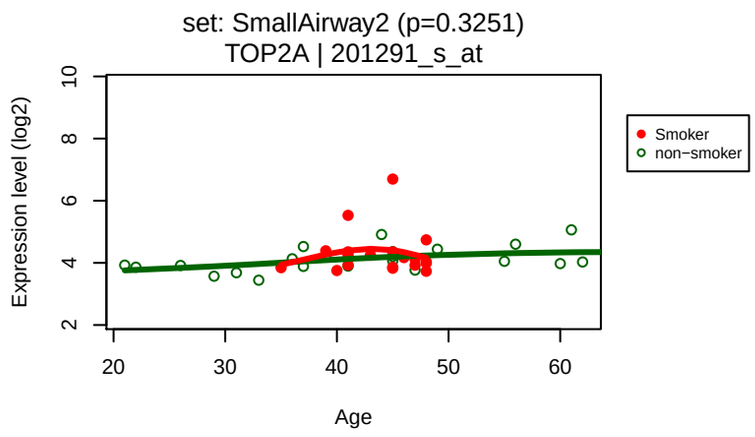
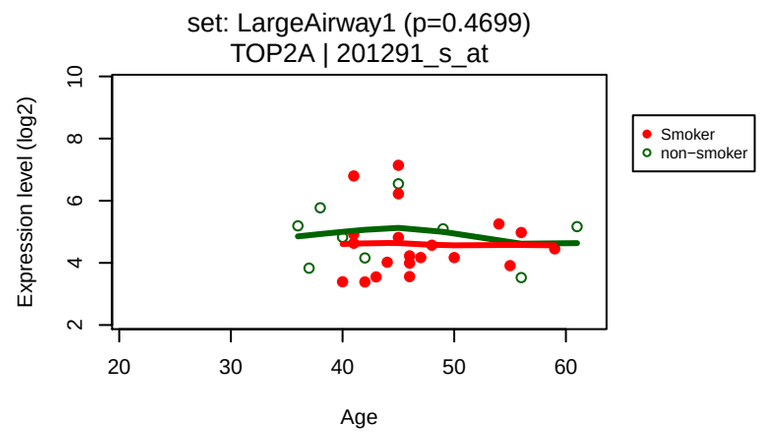

Gene expression of smokers and nonsmokers
gene: TOP2A | 201291_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3285
p=0.3251
p=0.0609
p=0.4699
p=0.6183
p=1e−04
p=0.5277
p=0.9444
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

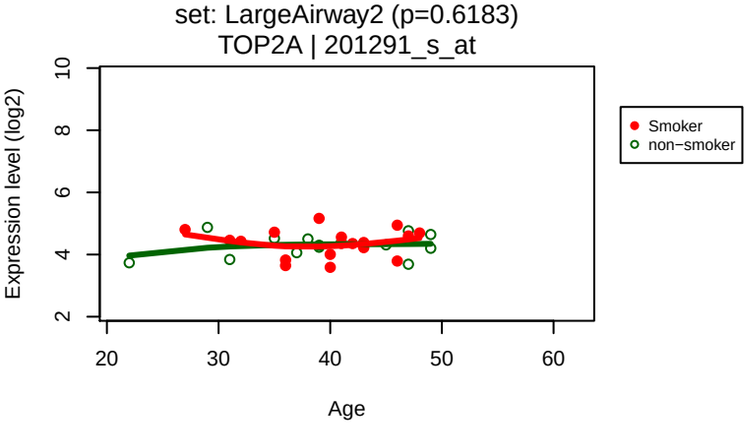

Gene expression ratio (smokers/nonsmokers)
gene: TOP2A | 201291_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3285
p=0.3251
p=0.0609
p=0.4699
p=0.6183
p=1e−04
p=0.5277
p=0.9444
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

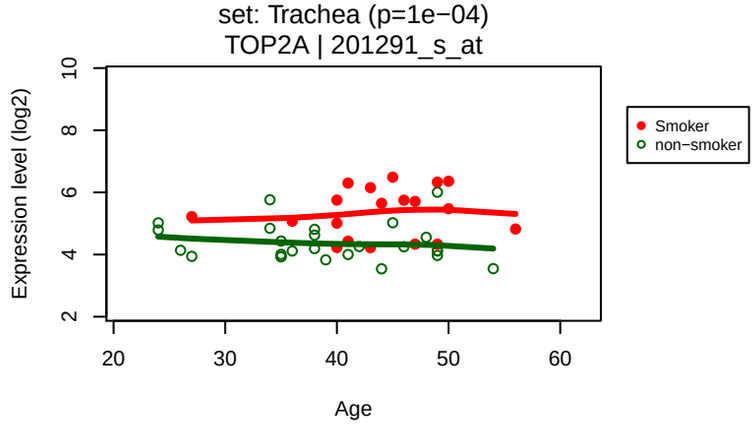

Gene expression ratio in Males
gene: TOP2A | 201291_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3981
p=0.2318
p=3e−04
p=0.1754
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TOP2A | 201291_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6439
p=0.505
p=0.0662
p=0.6346
SmallAirway2
LargeAirway2
Trachea
Macrophage2

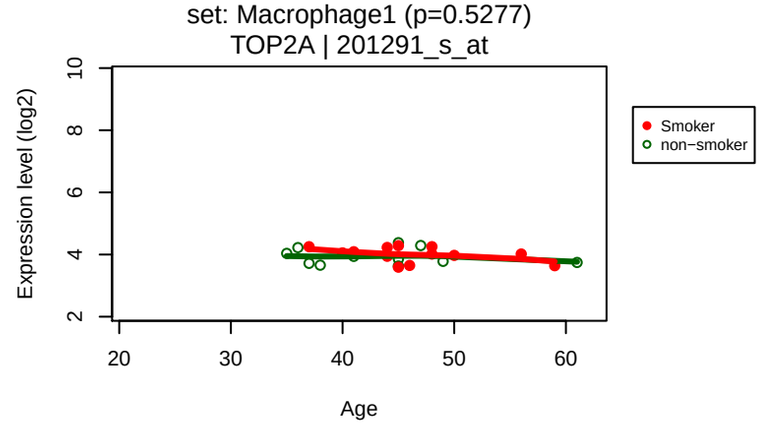

set: SmallAirway1 (p=0.3285)
TOP2A | 201291_s_at
Expression level (log2)
Age
Smoker
non−smoker

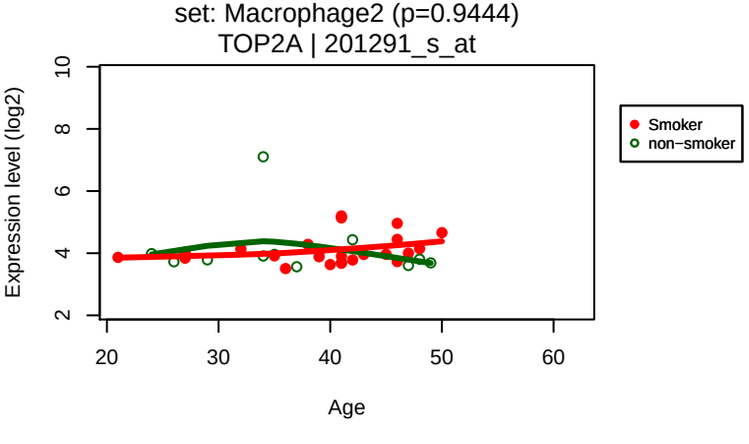

set: SmallAirway2 (p=0.3251)
TOP2A | 201291_s_at
Expression level (log2)
Age
Smoker
non−smoker

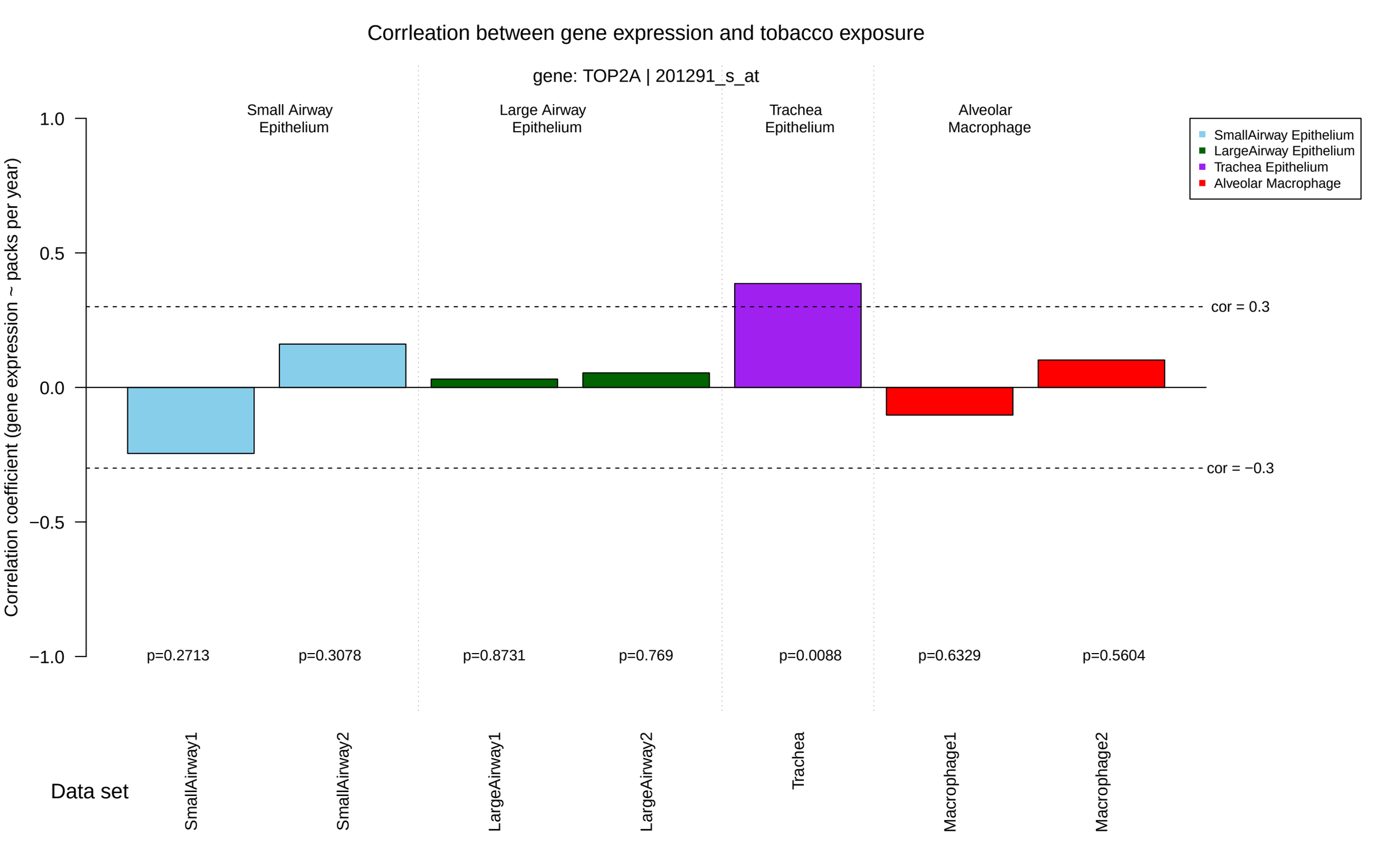

set: LargeAirway1 (p=0.4699)
TOP2A | 201291_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6183)
TOP2A | 201291_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=1e−04)
TOP2A | 201291_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5277)
TOP2A | 201291_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9444)
TOP2A | 201291_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TOP2A | 201291_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2713
p=0.3078
p=0.8731
p=0.769
p=0.0088
p=0.6329
p=0.5604
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2